

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.4 CELL DIVISION – FINAL EXPLANATION TO THE PHENOMENON

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Key question: How does the body produce the enormous number of new cells needed for growth? Think back to the baby photographs and the 14-year growth time-lapse from Lesson 1, where a newborn of about 3 kg became a teenager of 50 kg or more. Use what you observed and modelled in all ten lessons to write your Final Explanation. Answer each prompt in the space below it. Write in full sentences, use correct scientific terms, and support each claim with evidence from our investigations (diagrams, clay models, onion root tip squashes, animations, case studies). Compare your final answer with your first model from Lesson 1 at the end.

Part 1: Claim - Where do the new cells come from?

A newborn of about 3 kg becomes a teenager of 50 kg or more, yet we only eat food such as ugali, sukuma wiki and beans. Write a claim that answers the key question: where do the billions of new cells come from? In your answer, say what food does and does not do in growth. Then say how your thinking has changed since your Lesson 1 drawing.

Write your answer here:

Part 2: Evidence from mitosis - Growth and the right instructions

Use evidence from our diagram analysis, the clay model of 1 cell to 2 cells, and the onion (kitunguu) root tip squashes. (a) Describe what happens to the chromosomes before and during mitosis, naming the stages. (b) Explain how mitosis gives each new cell the right instructions. (c) In the root tip squashes, most cells were in interphase and fewer were dividing. What does this tell you about growing tissue? (d) Explain how repeated division (1, 2, 4, 8, 16...) can lead to billions of cells.

Write your answer here:

Part 3: Evidence from meiosis and other forms of cell division

(a) A sperm and an egg each come from a body cell with 46 chromosomes. Explain, using the calculation we did in Lesson 4, why meiosis is needed and how many chromosomes each gamete has. **(b)** Describe what meiosis produces compared with mitosis. **(c)** Name two ways a single organism can make new organisms or cells from one parent without sex cells, giving a living example of each, and say how they are similar.

Write your answer here:

Part 4: Why cell division matters

Answer the KICD inquiry question: How is cell division important? Use at least three examples from our case studies, including the healing cut, the maize or sukuma wiki plant, and tissue culture of bananas or tea in Kenya. For each example, state whether mitosis or meiosis is involved and why.

Write your answer here:

Part 5: Final Explanation - Claim, Evidence, Reasoning

Write your complete Final Explanation to answer the key question: How does the body produce the enormous number of new cells needed for growth? Use the Claim-Evidence-Reasoning structure. Include (1) your claim, (2) at least three pieces of evidence from our investigations (including mitosis and meiosis), and (3) reasoning that links the evidence to the growth of a child from 3 kg to 50 kg or more and to the wider importance of cell division.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the phenomenon	Gives a clear claim that new cells come from division of existing cells, explains that food gives energy and raw materials but does not become cells, and links directly to the 3 kg to 50 kg growth of a child and to the Lesson 1 model.	States that cell division makes new cells and links to the growth of a child, but the role of food or the link to the Lesson 1 model is brief or incomplete.	Claim is vague or incorrect (for example, food turns directly into cells or cells only grow bigger) and there is little link to the phenomenon.
Understanding of mitosis	Accurately describes DNA copying, the four stages and cytokinesis, explains how each daughter cell gets identical chromosomes and the right instructions, and explains doubling (1, 2, 4, 8, 16...) to billions of cells.	Describes mitosis and the formation of two identical cells with mostly correct stages, but explanation of how instructions are passed on or of doubling is partial.	Describes mitosis with major errors or omissions, such as missing stages or no mention of chromosomes being copied and shared.
Understanding of meiosis and other cell division (budding and	Explains with the $46 + 46 = 92$ reasoning why meiosis halves the chromosome	Correctly states that meiosis makes gametes with half the chromosomes and	Confuses mitosis and meiosis or does not correctly describe gametes, budding

binary fission)	number to 23, contrasts it correctly with mitosis, and accurately describes budding and binary fission with examples as asexual reproduction.	names budding and binary fission, but comparison or examples are limited.	or binary fission.
Use of evidence and Kenyan examples	Uses at least three specific pieces of evidence from the investigations (such as the onion root tip squashes, clay model, case studies, tissue culture of bananas or tea) and ties each to the claim.	Uses two relevant pieces of evidence, with some connection to the claim.	Uses little or no evidence, or gives general statements not taken from our investigations.
Reasoning, importance of cell division and scientific communication	Logically links evidence to the claim, explains the importance of cell division for growth, development, repair and reproduction, and uses correct terms in clear, well-organised Claim-Evidence-Reasoning writing.	Reasoning mostly links evidence to the claim and mentions some importance of cell division, with minor errors in terms or organisation.	Reasoning is missing or unclear, few uses of the importance of cell division, and scientific terms are used incorrectly or not at all.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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